

Ahmed El-Hajjar

Technical Writer, Software Engineer

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Leadership and Initiatives

Ottawa Computing Group – [meetup.com/ottawa-computing-group](https://www.meetup.com/ottawa-computing-group)

Founder and Host • 300+ events • 1300+ members • 6+ years

June 2019 – Current

Weekly in-person social club spanning technology, current affairs, project showcases and free-form discussions.

- Plan, coordinate and moderate one of the few technology meetups that remained fully in-person after the pandemic.
- Attendees share projects, challenge each other's perspectives, and occasionally discover job opportunities or collaborators.
- No presentations, agendas, speakers, or predetermined topics. Conversations are entirely driven by attendees.
- Welcomes everyone from software developers, blue-collar experts, content creators, entrepreneurs and even non-techies.

Selected Portfolio Projects

PoetWrite – github.com/cdahmedeh/PoetWrite

Technical Writer, Software Engineer, Architect

Solo Open-Source, Early Development, Documentation Focused, Java

Research-phase word processor for poets featuring rhetorical analysis and lexicographic assistance. Fully documented software architecture and analyses engines.

- Comprehensively documented custom asynchronous MVVM UI framework using reactive patterns, bespoke caching systems and a clever lazy-loading dependency-injected services.
- Research how far deterministic, AI-free language analysis and assistance can be pushed before requiring machine learning techniques.
- Sole ownership of project using software engineering practices. Such as architecture design, UX design, software development, documentation, quality assurance and TDD-driven testing.

Crosswind – github.com/cdahmedeh/Crosswind

Technical Writer, Reverse-Engineer, Programmer

Solo Open-Source, Prototype Release, .NET C#

Reverse-engineering effort to create compatibility layer that bridges modern flight planning software to decades-old legacy flight simulators.

- Crosswind allows users of older flight simulators to use modern flight planning tools that don't have published APIs for providing flight or environment data or access to aircraft analytics.
- Clean-room reversed-engineered undocumented protocols using packet sniffers and injected API monitors. To develop a data translation layer between legacy simulator interfaces (FSUIPC) and modern flight-planning APIs (Navigraph Simlink).
- Interfaces and protocol findings documented in source code along with data structures used by said protocols.
- Distributed as a complete open-source deliverable. With release notes, compatibility lists, build instructions, usage instructions in the repository home page. Included is a fully compiled distributable with changelog, download links and hashes for validation.

TopRoms – github.com/cdahmedeh/TopRoms

Technical Writer, Curator, Infrastructure • 176 GitHub Stars • 4000+ titles • 60+ platforms • Since 2017 Solo, Fully Released

Curated preservation project focused on historically significant and high-quality retro games. Designed as a more practical alternative to full-archival collections.

- Research-driven curation methodology combining review aggregation, historical sources, community feedback.
- Authored extensive project documentation covering collection history, curation methodology, emulator recommendations, platform compatibility, release notes, and setup guidance.

- Built and maintain a large-scale ROM content-delivery project end-to-end. Maintaining hosting and distribution infrastructure as a seedbox. Conversion of formats for emulator compatibility using tools from the MAME project.

Selected Publications

Automatic Transmission Simulation in Games – automatic.cdahmedeh.net

Technical deep-dive comparing accuracy of automatic transmission simulation across driving games.

- Results from research project investigating why the majority of driving games have inaccurate modeling of automatic transmissions. And describe, test and demonstrate findings of rare and obscure titles that accurately reproduced the behaviour of automatics found in road cars.
- Explains the engineering principles behind automatic transmission behaviour in a manner accessible to driving game enthusiasts.
- Develop a testing methodology to identify the various distinctive characteristics and features for the in-game analyses.
- Includes comparative testing and video demonstrations documenting these tests across simulators and driving games.
- Widely referenced by driving simulation enthusiasts and developers as a resource on transmission modeling.

The Sad Demise of Propulsion Controlled Aircraft – pca.cdahmedeh.net

Storytelling and historical technical case study of propulsion-controlled aircraft and its impact on aviation safety and its eventual cancellation and abandonment.

- Researched a little-known NASA project that allowed pilots to control and land aircraft with complete hydraulic and flight control failure using engine thrust alone.
- Connected historical accident reports, NASA and McDonnell-Douglas flight-test programs, aviation safety research, and simulator demonstrations through storytelling.
- Supported with videos of CVR recordings, accident recordings, flight-test program initiatives and results, and reference to a simulator demo demonstrating throttle-only flight control.
- Reflection on how FAA considerations influenced the rejection of a technology that may have prevented several major aviation disasters.

Professional Experience

Flybits

Technical Writer, Solutions Team

March 2024 – June 2026

Owner and author of documentation ecosystem for complex fintech platform that had major fragmentation knowledge challenges. Serving engineering and developer documentation, comprehensive product user guides, cross-departmental knowledge transfer, support enablement and professional services.

Creating a from-scratch comprehensive SDK development reference documentation and tutorials from fragmented sources.

- Flybits was a customer-engagement platform where marketers created and managed personalized content experiences through a web application, while the SDK delivered those experiences to mobile and web applications.
- SDK knowledge was fragmented across outdated documentation, engineering tribal knowledge, support teams, and professional services.
- Lack of accurate reference material increased support burden and made developer onboarding difficult.
- Collected information through SME interviews and knowledge-transfer sessions with engineering, product, solutions, and professional services teams.
- Supplemented code examples and snippets through SDK source-code analysis, internal demo applications, and reverse-engineering of undocumented library functions.
- Developed prototype applications in Swift and Kotlin to validate documentation accuracy and generate executable code samples, allowing documentation to be written from direct implementation experience rather than theory alone.

- Designed documentation structure for searchability and discoverability through SEO-like search optimization and metadata tagging.
- The majority of the documentation was written from scratch via the GitBook platform. Used a combination of reference-style and step-by-step writing patterns tailored to both novice and experienced developers.
- Structured documentation into a progressive learning path combining tutorials, implementation guides, troubleshooting references, API configuration guides, and platform-specific Android and iOS documentation.
- Developed and maintained a multi-hundred-page SDK documentation system covering the full developer lifecycle from initial setup and authentication through testing, deployment, and advanced platform capabilities.
- Kept and maintained older documentation for legacy and posterity, as some customers were still transitioning to newer frameworks.

Designing and composing User Guide for primary web application platform for client end-users and marketers.

- Flybits operated a web application, called Experience Studio, used to design and manage personalized content experiences across mobile and web applications, including contextual content presentation, audience targeting, scheduling, prioritization, and analytics.
- Primary audiences had limited technical backgrounds, including marketers and product owners. Documentation balanced ease of use for business stakeholders with the technical accuracy required for engineering collaboration.
- Platform development was fast-paced, with major UX and workflow changes occurring frequently throughout the product's evolution and redesign efforts.
- Virtually no documentation existed for the redesigned platform, with available material limited to deprecated documentation from earlier generations of the product.
- Developed a strategy for creating a comprehensive user guide from scratch, working closely with product, support, and solutions teams to ensure documentation accurately reflected platform behavior and customer needs.
- Documentation structure was inspired by large-scale enterprise software user guides and the progressive learning approach found in flight operating manuals.
- Created a gradual learning path that introduced concepts in concise stages before progressing to workflows, tutorials, and practical examples, rather than relying primarily on reference-style paradigms.
- Support and solutions teams reported substantially less time spent explaining platform fundamentals, allowing greater focus on developing customer-specific solutions and implementation guidance.
- Recognized that many end-users and marketers are reluctant to read documentation, leading to a simplified structure with extensive visual aids, animations, strong searchability, and step-by-step explanations and navigation.
- Implemented a tagging system to track documentation coverage of platform features, as a management strategy as the platform evolved and new features were implemented.

Writing onboarding documentation for finance and accounting team.

- Flybits underwent a major transformation of its finance operations, including billing, reimbursement, payroll, and accounting workflows, accompanied by migration to entirely new platforms and processes.
- The redesigned processes were primarily understood by the accounting leads, who required comprehensive onboarding documentation to train and support the rest of the finance team.
- Had no prior experience with finance or accounting systems and was required to rapidly learn unfamiliar concepts and terminology under tight project timelines due to the urgency of the documentation effort.
- Structured documentation around financial activities and workflows rather than linear beginning-to-end narration, allowing team members to learn and reference individual responsibilities independently.
- Authored documentation in Google Docs rather than developer-oriented documentation platforms to align with the limited technical-familiarity and working practices of the finance team.
- Due to confidentiality restrictions, direct access to financial systems was unavailable. Documentation was created through detailed note-taking, meeting transcriptions, process mapping, copious screenshots, and frequent and long SME interviews.

Syntronic

Technical Writer, Business Analyst

May 2022 – November 2023

Requirements analyst and technical communicator for a variety of research initiatives, including medical labs, embedded systems and software development.

Requirements analysis contributor for an automated medical sterilization lab.

- Conducted client interviews and requirements-gathering sessions to determine the operational and technical needs of a newly developed automated medical sterilization laboratory.
- Analyzed, validated, and refined requirements by identifying ambiguities, contradictions, missing information, and implementation risks through collaboration with clients, analysts, and subject-matter experts.
- Transformed business and operational requirements into software specifications, implementation plans, and development roadmaps for future software engineering initiatives.
- Authored user stories, acceptance criteria, and test scenarios to support development and quality-assurance activities.
- Collaborated with developers and UX specialists to formalize workflows, use cases, and functional requirements prior to implementation.
- Contributed to business-analysis reports documenting project objectives, development plans, assumptions, risks, and findings discovered throughout the requirements-engineering process.

Automating requirements management with continuous integration and deployment platforms.

- Requirements, specifications, and implementation plans were maintained primarily in Microsoft Word and PDF documents, resulting in significant manual effort to transfer information into Azure DevOps.
- Designed and developed a cross-platform Java application to automatically extract requirements, user stories and acceptance criteria from specification documents.
- Integrated with Azure DevOps APIs to automatically synchronize development work items to eliminate the need for manual data entry thus improving consistency between specifications and project tracking.

Technical communications for emerging cloud and embedded technologies.

- Authored whitepapers, brochures and marketing material describing digital-transformation initiatives, and embedded-system products for prospective clients.

Early Career

Amazon, Entrust, 1VALET

Technical Writer, Consultant

October 2020 – October 2022

Knowledge-transfer specialist focused on complex technical systems, onboarding, and undocumented functionality.

Cisco, ESDC Service Canada, Ford Motor Company

Software Consultant

February 2019 – October 2020

Architecture, deployment, automation, and enterprise modernization initiatives.

Lixar, NorthStar, ShopLiftr

Software Developer

December 2013 – February 2019

Full-stack developer specializing in APIs, cloud platforms, emerging technologies, and software architecture.

Additional Experience

Uber

Partner and Driver • Part-Time • 4.9 Star Rating • 1000+ trips

January 2019 – September 2025

Independent customer-service and transportation operation serving riders throughout Ottawa-Gatineau while managing scheduling, bookkeeping, and day-to-day business operations. Helped develop communication and interpersonal skills that could be applied in other areas in my field.

Volunteering

Ottawa Stray Cat Rescue

Lead Volunteer • Occasional

February 2020 – Current

Organizing fundraisers and adoption events for the local animal rescue for stray and feral cats throughout Ottawa.

Selected Essays

Electric Youth -- Marnie – marnie.cdahmedeh.net

Analytical essay exploring music theory, lyrical themes, visual storytelling, and artistic intent within a contemporary synth-pop composition.

Education

Studies in Software Engineering, Computer Engineering, Computer Science

University of Ottawa

September 2009 – December 2014

Studies in Technical Communications

Simon Fraser University (Online)

June 2018 – November 2018

Keywords (Application Tracking System Friendly)

Documentation

Specifications, Requirements, Technical Documentation, API Documentation, Knowledge Transfer, User Guides, Success Guides, User Experience, User Stories, Deployment Documentation, Support Documentation, Bug Reports, Test Cases, ReadMe, Version History, Changelogs, Prototyping, SDK Documentation, Mobile Documentation, Developer Documentation

Content Creation

Whitepapers, Case Studies, Ghostwriting, Copywriting, Editing, Blogging, Marketing, Social Media, Video Editing, Image Authoring, Announcements, Advanced Research, Fact Checking

Soft Skills

Project Management, Servant Leadership, Mentoring, Peer Reviews, Interviews, Interdepartmental Communication, Subject Matter Expert (SME) Solicitation, Knowledge Transfer, Fragmentation Consolidation, Event Hosting, Moderation, Social Club

Business Analysis

Requirements Gathering, Specifications Development, Customer Interviews, Business Analysis Summaries, Visualization, Project Management, Product Ownership, Code Exploration

Visual Modeling

Flowcharts, Processes, State Diagrams, Mindmaps, UML, Sequence Diagrams, Swim Lanes

Development Methodologies

Agile, SCRUM, Waterfall, KANBAN, Kaizen, GTD, Pair Programming, Code Reviews, Grooming Sessions

Task Tracking

JIRA, Trello, Todoist, PVCS, Rally

Publishing Tools

Microsoft Word, Microsoft Publisher, LibreOffice Writer, Adobe InDesign, Scrivener, Google Docs, Microsoft Visio, draw.io, Balsamiq, Antidote, Markdown

Authoring Tools

KDenLive, Sony Vegas, Adobe Photoshop, GIMP, FL Studio, Renoise, DaVinci Resolve

Collaborative Platforms

SharePoint, Google Drive, Quip, Confluence, MediaWiki, GitBook, GitHub

Website Management Systems

WordPress, Squarespace, Wix

Spoken Languages

English, French, Arabic

Translation

French to English

Programming Languages

Java, Visual Basic, C#, C++, Python, Ruby, Go, Pascal, Prolog, Scheme, PHP, Perl, Lua, Informix, Swift, Kotlin

Programming Styles and Architectures

Procedural, Object Oriented, Lambda, Functional, Markup, Query Languages, MVC, MVVM, Server-Side Rendering, Single Page Applications, Multi-Threading, Caching, Dependency Injection, Advanced Data Structures, Annotations, Annotation Processing, Test-Driven Development, Reverse-Engineering, Reactive Programming, Asynchronous

Web Languages

HTML, CSS, JavaScript, vue.js, Angular, React

Shell Languages

bash, zsh, cmd.exe, Windows Batch, PowerShell

Build Systems and Tools

Ant, Maven, Gradle

Integrated Development Environments

Eclipse, Android Studio, IntelliJ IDEA, Visual Studio Code, SharpDevelop, Lazarus, Rider, Resharper, Notepad++, vim

Version Control

git, CVS, SVN

Developer Platforms

GitHub, GitLab, Bitbucket, Azure DevOps

Continuous Integration and Deployment

Jenkins, TeamCity, Bamboo, Azure DevOps

Desktop Frameworks

Swing, SWT, JFace, JavaFX, JIDE, JGoodies, WebLaF, FlatLaf

Mobile Development

Android, Firebase, APNs, Retrofit, ORMLite, Prototyping

Backend Frameworks

Java EE, Spring, Sprint Boot, JHipster, Django, Ruby on Rails,

Web Frameworks

Angular, Ruby on Rails, JSP, JSF, PrimeFaces

Application Programming Interfaces

REST, SOAP, RMI, Swagger, Postman, flood.io

Serialization

JSON, YAML, XML, DTO

Database, ORM and Migration

SQL, NoSQL, PostgreSQL, MySQL, SQL Server, Informix, Oracle, JDBC, HikariCP, JPA, Hibernate, EclipseLink, MyBatis, Liquibase, Flyway, H2, SQLite

Testing

JUnit, jMock, Mockito, Selenium

Interop

JNA, Com4J, C# Interop

Generic Libraries

Apache Commons, Guava, Retrofit, Project Lombok, JodaTime, JSR-310, Log4j

Parsing

ANTLR, Regular Expressions (regex), grep

Public APIs

Microsoft Cognitive Services, GitHub, JIRA, Google Maps, Amazon AWS, Reddit, RSS

Services

samba, NFS, qBittorrent, Syncthing, Postgres, Apache, nginx, JBoss, WildFly, H2, RDP, VNC, SteamCMD, mpd, ShoutCast, systemd, Multichain, Asterisk, SIP, PBX, Tomcat, IIS

Operating Systems and Administration

Windows, macOS, Linux (Ubuntu, Debian, Arch Linux, Gentoo, RedHat, OpenSUSE, SLES, EndeavourOS), Package Managers, GNU Utilities, SSH, RDP, Windows Server, Tiling Window Managers, Rooting, Jailbreaking, JTAG, BIOS, EFI, Disk Encryption, Full-Image Backups

Virtualization and Containers

VirtualBox, VMWare, Docker, Docker Compose, Appliance, Windows Subsystem for Linux

Cloud

Amazon AWS, Microsoft Azure, DigitalOcean, Linode, Airtime, Namecheap, voip.ms, CloudWatch, flood.io, S3, SageMaker, Amazon ML, Redshift, Amazon DataPipeline, DynamoDB, LogEntries, HTTPS Certification

Technologies

Machine Learning, Artificial Intelligence, Blockchain, Optical Character Recognition, Facial Recognition, Bluetooth, Proximity, Crash Detection, Software Radios, Embedded

Electronics

Soldering, Circuitry, PCB Components, Arduino, Teensy, Raspberry Pi, Fritzing, Autodesk Fusion, 3D Printing, Basic FPGA Design, Quartus

Automotive

AllData, Basic Car Maintenance, Manual Transmissions, Rev-Matching, Double-Clutching, Heal-Toe, Complex Navigation

Simulation

Microsoft Flight Simulator, X-Plane, BeamNG.drive, Automation, Orbiter

Research

Non-Academic Research, Case Studies, Reviews, Music Analysis, Blogging, Content Curation